# MIZORAM PUBLIC SERVICE COMMISSION

## COMPETITIVE EXAMINATION FOR RECRUITMENT TO THE POST OF ASSISTANT CONTROLLER OF MINES UNDER GEOLOGY & MINING, COMMERCE & INDUSTRIES DEPARTMENT, GOVERNMENT OF MIZORAM.

## (MARCH-2026)

## TECHNICAL PAPER-II

*(Use of Scientific calculator is permitted.)*

Time Allowed : 2 hours FM : 200

***PART - A (Multiple Choice Questions) (100 Marks)***

*All questions carry equal mark of 2 each. Attempt all questions.*

*This Section should be answered only on the **OMR Response Sheet** provided.*

**1.** An air quality parameter required to be monitored under the Indian National Ambient Air Quality Standards is:

(a) As (b) Pb

(c) Hg (d) Silica

**2.** The value of COD is related to BOD as:

(a) COD > BOD (b) COD = BOD

(c) COD < BOD (d) There is no relation

**3.** The limit of explosibility of a mixture of combustible gases can be computed by which of the following equation?

(a) Pythagoras equation (b) Le Chatelier equation

(c) Simon's equation (d) Solmon's equation

**4.** What is the minimum percentage of $O_2$ to make methane air mixture explosive?

(a) 8% (b) 10%

(c) 12% (d) 15%

**5.** Gas chromatography separates & analyze components of gas mixture by:

(a) Adsorbent (b) Molecular Species

(c) Boiling Water (d) Catalyst

**6.** As per Indian law the methane concentration should not exceed "A%" in any part of the mine. A is given by:

(a) 0.5 % (b) 0.75 %

(c) 1 % (d) 1.25%

**7.** What is the basic principle of chain surveying?

(a) Dialing
(b) Chaining
(c) Traversing
(d) Triangulation

**8.** Match the following:

| **Instruments** | **Purpose/Measurement** |
|---|---|
| 1: Abney's level | a: Horizontal and vertical angles |
| 2: Pantograph | b: Area of plotted figure |
| 3: Planimeter | c: Enlargement and reduction of plotted maps |
| 4: Box sextant | d: Angle of inclination |

(a) 1-a, 2-c, 3-d, 4-b
(b) 1-c, 2-b, 3-d, 4-a
(c) 1-d, 2-a, 3-d, 4-c
(d) 1-d, 2-c, 3-b, 4-a

**9.** Which is used when leveling of the highest precision is required?

(a) Telescope
(b) Invar staff
(c) Tape
(d) Object glass

**10.** Name the method which is particularly suitable when a contoured map of hill is required:

(a) By squares method
(b) Tacheometric method
(c) Cross-section
(d) Indirect method

**11.** It is a simple device attached to the wheel of any vehicle and registers the number of revolutions of the wheel. Knowing the circumference of the wheel, the distance traversed may be obtained:

(a) Passometer
(b) Pedometer
(c) Speedometer
(d) Odometer

**12.** What indication do we get when contour lines lie closer?

(a) It means that slope of land is gentle
(b) It means that slope of land is steep
(c) It means that the land is flat
(d) It does not give any indication

**13.** A 6×19 wire rope would be:

(a) 6 inches in diameter and 19 fathoms long.
(b) 6 inches in circumference with 19 strands.
(c) 6 strands with 19 wires in each strand.
(d) 19 strands with 6 wires in each strand.

**14.** In a drum hoisting system through a vertical shaft, overwinding is prevented by:

(a) Lilly controller
(b) Detaching hook
(c) Caliper Brake
(d) Safety catch

**15.** The sequence of attachments between the winding rope and the cage is:

(a) bull chains, triangular plate, capel, detaching hook
(b) detaching hook, bull chains, triangular plate, capel
(c) triangular plate, bull chains, capel, detaching hook
(d) capel, detaching hook, bull chains, triangular plate

**16.** Main advantage of using main and tail rope haulage is:

(a) No derailment of tub occurs.
(b) Suitable for high output.
(c) Suitable for undulating roadways and one operator can drive both drums.
(d) Long train of tubs can be attached.

**17.** In relation to pump, head is known as:

(a) Distance between pump motor and discharge point.

(b) Vertical distance between suction end to discharge point.

(c) Distance between suction end to pump impeller.

(d) Pump impeller to discharge point

**18.** Match the following:

| **Blast Problem** | **Cause** |
|---|---|
| P: Misfire | 1: Poor stemming |
| Q: Vibration | 2: Low current |
| R: Blown-out shot | 3: Excess charge |
| S: Cut-off shot | 4: Improper delays |

(a) P- 3, Q- 2, R - 4, S-1

(b) P- 4, Q-1, R - 2, S-3

(c) P- 2, Q- 3, R - 1, S-4

(d) P- 1, Q-2, R - 4, S-3

**19.** No drill shall be used for boring a shot hole unless it allows a clearance of at least "A" cm over the diameter of the explosive cartridge. "A" is given by:

(a) 0.5

(b) 1

(c) 2

(d) 0.3

**20.** No shot shall be fired unless the length of fuse is at least "A" meter. A is given by:

(a) 1.2 m

(b) 1.8 m

(c) 2.5 m

(d) 4 m

**21.** Pop shooting or plaster shooting is a type of:

(a) Primary blasting

(b) Secondary blasting

(c) Muffling

(d) Clay capping

**22.** During drilling, the hole gets filled with cuttings and has to be cleaned out frequently. This is done with the help of:

(a) Derrick

(b) Calyx shot drilling

(c) Sludger

(d) V-chisel

**23.** Which of the following is the oxidizing agent in an explosive?

(a) Charcoal

(b) Aluminum

(c) Ammonium nitrate

(d) Magnesium carbonate

**24.** Which of the following has poor water-resistant property?

(a) ANFO

(b) Slurry explosive

(c) Emulsion explosive

(d) NG based explosive

**25.** Which of the following prime characteristic the NONEL blasting has?

(a) It has greater shock to the surrounding.

(b) It is noiseless in character.

(c) It has tremendous effect on explosive column.

(d) It cannot be used in a place where there is a chance of stray current.

**26.** In ambient air of industrial area, the permissible concentration of suspended particulate matter ($PM_{10}$) measured on 24 hours average in $\mu g/m^3$ is:

(a) 100 (b) 200

(c) 300 (d) 400

**27.** Light wave travel with a velocity of:

(a) $3 \times 10^{10}$ cm/s (b) $3 \times 10^{11}$ cm/s

(c) $3 \times 10^{12}$ cm/s (d) $3 \times 10^{13}$ cm/s

**28.** The most important parameter of a dust that governs its physiological effect is:

(a) Concentration of dust in air (b) Shape of dust

(c) Time of exposure (d) Size of dust

**29.** Dust generation in drilling maybe reduced by using:

(a) Low RPM (b) Sharp bit

(c) High RPM (d) Low thrust

**30.** _____ disease is caused by working in places of insufficient light.

(a) Pneumoconiosis (b) Ancylostomiasis

(c) Nystagmus (d) Silicosis

**31.** The noise level at 10 m from a long conveyor belt is 95 dBA. What is the noise level at 100 m?

(a) 80 dBA (b) 85 dBA

(c) 90 dBA (d) 95 dBA

**32.** The permissible value of oil and grease discharge from the mine workshop is:

(a) 120 mg/L (b) 150 mg/L

(c) 220 mg/L (d) 250 mg/L

**33.** The combined effect of curvature and refraction in levelling is an error which is:

(a) Additive (b) Subtractive

(c) Multiplicative (d) Divisive

**34.** The method of leveling adopted to determine the difference of level between two points at a considerable distance apart with great precision is:

(a) Hypsometry (b) Profile levelling

(c) Reciprocal levelling (d) Trigonometrical levelling

**35.** The fore bearing of a line AB is 136°. What will be the back bearing of the line AB?

(a) 314° (b) 315°

(c) 316° (d) 317°

**36.** GIS stands for:

(a) Global Information System (b) Geological Information System

(c) Geographical Information System (d) Group Information system

**37.** Rope used for winding purpose is:

(a) Lang's lay rope (b) Ordinary lay type

(c) Fully locked coiled rope (d) Half locked coiled rope

**38.** Minimum gradient for Gravity Haulage operation is:

(a) 13° (b) 14°

(c) 15° (d) 16°

**39.** Tractive effort of a locomotive is given by:

(a) $\frac{W}{\mu}$ (b) μ. W

(c) W - μ (d) W + μ

**40.** Compared to conventional belt conveyors, high angle conveyors are suitable for:

(a) Long horizontal distance only (b) Very low speed operation

(c) Steep gradient transport (d) Fine material handling only

**41.** Brake Horse Power (BHP) of a pump is given by:

(a) $\frac{Q * H_T}{3960}$ (b) $\frac{Q * H_T * Sp}{3960 * \mu}$

(c) $\frac{Q * H_T * \mu}{3960}$ (d) $\frac{3960}{\mu * H_T}$

**42.** If a pump develops 10 Bar of pressure and has a flow rate of 1500 L/min, what will be the pump motor power at 90% efficiency?

(a) 26 kW (b) 28 kW

(c) 30 kW (d) 32 kW

**43.** The maximum permissible angle of fleet should not exceed:

(a) 1 (b) 1.5

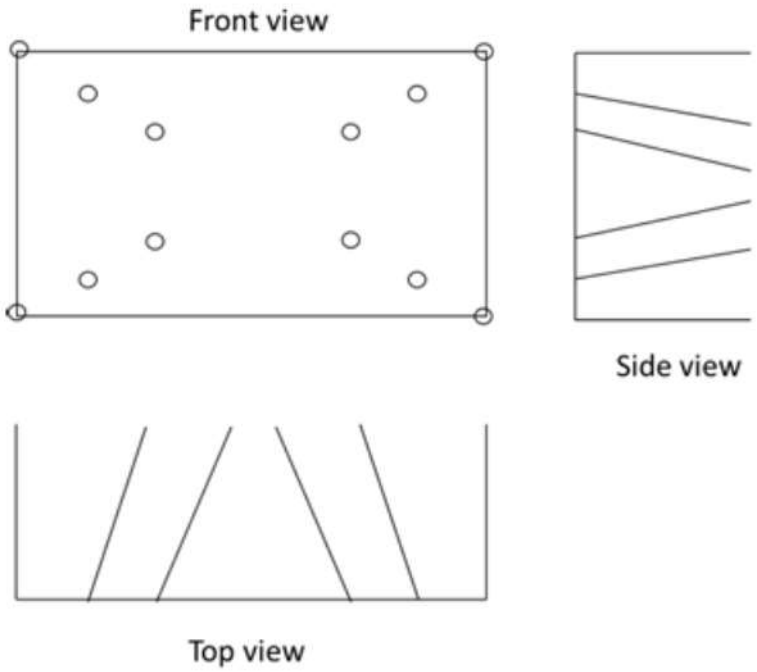
(c) 2 (d) 2.5

**44.** Detonating chord has a VOD of:

(a) 4500-5000 m/s (b) 5500-6000 m/s

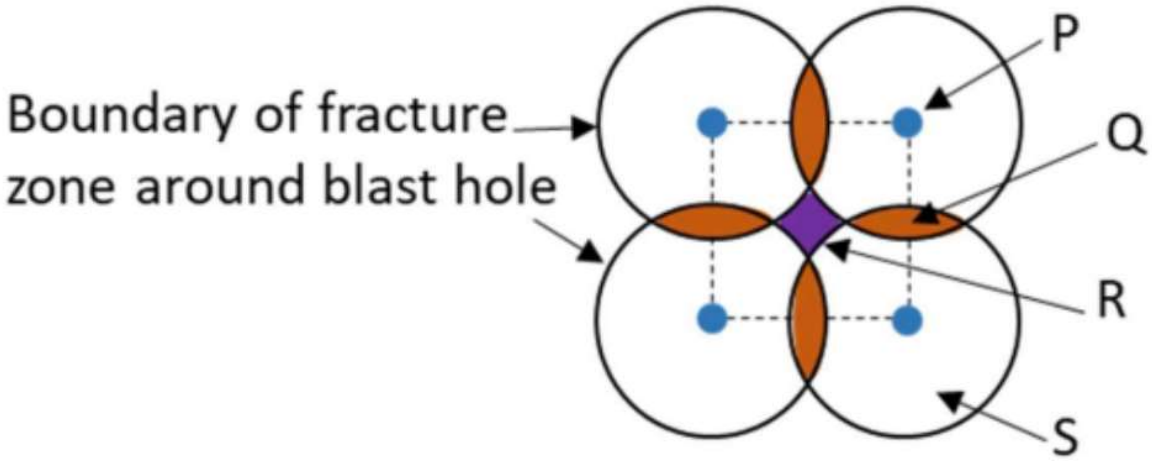
(c) 6500-7000 m/s (d) 7500-8000 m/s

**45.** The blast pattern of a coal face shown in the figure represents:




(a) Burn Cut
(b) Pyramid cut
(c) Wedge cut
(d) Drag cut

**46.** A square pattern of blasting is shown in the figure. For the case of simultaneous blast, identify the zone of no fragmentation.




(a) P
(b) Q
(c) R
(d) S

**47.** To prevent slope failure the blasting system selected should be:

(a) Pre-split
(b) Muffled
(c) Deck charging
(d) All of the above

**48.** In case of opencast mine blasting, the danger zone comprises of area within the diameter of:

(a) 400 m
(b) 500 m
(c) 600 m
(d) 300 m

**49.** At zero oxygen balance, the ratio of oxidizers to fuel ratio for ANFO is:

(a) 92 : 8 (b) 93 : 7

(c) 94 : 6 (d) 95 : 5

**50.** A coal heading 4m wide and 2.5 m high has an advance of 1 m per blasting round. The amount of explosive used in the blasting round is 6 kg. If the sp.gr of the coal is 1.5. the powder factor is:

(a) 2.0 te/kg (b) 2.5 te/kg

(c) 3.0 te/kg (d) 3.5 te/kg

## PART- B *(Conventional Type) (100 Marks)*

*This Section should be answered only on the **Answer Sheet** provided.*
*Marks for each question is indicated against it. Attempt all questions.*

**1.** The noise levels measured during the operation of three machines are 80 dBA, 75 dBA and 85 dBA respectively. Determine the equivalent noise level for the above three machines. **(5)**

**2.** The sound pressure level is measured as $5 \times 10^{-4}$ N/m$^2$. What is the noise level in dB? **(5)**

**3.** Define the following in one or two sentences. Also give the mathematical expression (formula) for each. **(5×1=5)**

(i) Correction Factor due to incorrect chain.

(ii) Correction Factor for slope.

(iii) Correction Factor for temperature.

(iv) Correction Factor for pull.

(v) Correction Factor for sag.

**4.** The ground distance between the mine manager's office and pit top is 300 m and it is represented by 3 cm in the mine plan. What is the representative factor? **(5)**

**5.** What is Friction / Koepe Winding? Write the mathematical expression governing the relationship between the tensions on the two sides of the rope as it passes over the friction wheel. **(5)**

**6.** It is proposed to connect two straights of a road by a simple circular curve.

If the maximum speed of the vehicle is 60 km/h and the centrifugal ratio for the road is ¼, calculate the minimum radius of the curve in metre. **(5)**

**7.** A 1100 V, 3 Φ power supply system of a mine draws a load of 185 kW. The ammeter reading shows 115 A. Calculate the power factor of the system. **(5)**

**8.** Define explosive in the context of Mining Engineering. What are the two main classification of explosive? Give one example each. **(5)**

**9.** A coal heading 4m wide and 2.5 metre high has an advance of 1 metre per cycle. The amount of explosive used in blasting is 6 kg. Taking specific gravity of coal as 1.5, calculate the powder factor. **(5)**

**10.** Briefly discuss the main ingredients of an explosive, namely: the Oxidizer, Fuel, and Sensitizer. **(5)**

**11.** Write a short note on sources of air pollution in mining industries. **(4)**

**12.** Explain the major occupational diseases associated with mining and discuss the different types of dust hazards that lead to pneumoconiosis. **(5)**

**13.** Explain in brief what is Bio-chemical Oxygen Demand (BOD) and Chemical Oxygen Demand (COD). **(5)**

**14.** Explain five (5) measurement parameters of Levelling. **(5)**

**15.** Explain the working principle of GPS. **(5)**

**16.** Explain in brief the different types of Conveyor Transport used in Mines. **(5)**

**17.** Explain the factors affecting the selection of a mine pump and define the terms capacity and head. **(5)**

**18.** Explain the concept of ground vibration due to blasting. Discuss the methods of its control and the DGMS permissible limits of Peak Particle Velocity (PPV) for different structures. **(6)**

**19.** What is Powder Factor? Discuss the factors affecting it. Also state the maximum permissible charge per shot hole as per DGMS regulations. **(5)**

**20.** A blast consists of 8 delay circuits with 30 ms interval between delays. Each hole contains 4 kg of explosive and two holes fire per delay. The blasting is carried out at a distance of 200 m from a domestic house (kacha brick & cement). **(5)**

The ground vibration predictor equation is:

$$V = K\left(\frac{D}{\sqrt{Q}}\right)^{-1.6}$$

Where:

K= 1000

D = distance in metres

Q = maximum charge per delay in kg

i) Calculate the maximum charge per delay.

ii) Calculate the Peak Particle Velocity (PPV).

iii) Comment whether the blast is within DGMS permissible limit (8–25 Hz range).

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